Name:____ Roll Number:__________________

Quiz-3

Max points: 15 **Max Time: 15 mins**

Q.1. [6 points]

Mark True (T) or False (F), fill in the blanks, or choose the correct choice for the statements below.

1. Entropy of a dataset having three classes and equal number of instances of each class is:

(A) 0 (B) 1 (C) 0.82 (D) 1.58

2. Approximate inductive bias of a decision tree is:

(A) All the attributes in a dataset must be selected (B) Deeper trees are preferred

(C) Shallower trees are preferred (D) None of the given options

3. To incorporate continuous attributes in a decision tree algorithm, we can discretize them so that the information gain is maximized.

(A) True (B) False

4. Bootstrapping in random forests means that:

(A) Creating multiple boosted datasets from the original one sampled without replacement

(B) Creating multiple boosted datasets from the original one sampled with replacement

5. All features of a classification problem must be used in random forests.

(A) True (B) False

6. Decision trees have lower variance than the random forests.

(A) True (B) False

Q.2. [5+3+1=9 points]

a) Gain ratio should be used instead of information gain in decision trees. Why?

Ans:

Attributes with many values bias selection in their favour if information gain alone is used. To undo this effect, we may use gain ratio, which penalizes the information gain by the splitting tendency of such attributes. Splitting tendency is measured the same way as the information gain but with respect to the attribute values instead of the target classification.

Name:____ Roll Number:__________________

b) Name two post-pruning techniques for decision trees induced by the ID3 algorithm. Explain how reduced error pruning is done.

Ans: The two post-pruning techniques are:

1. Reduced error pruning
2. Rule post-pruning

Reduced error pruning is done considering removal of a non-leaf node and replacing it with the class label (leaf-node) of the majority of samples at that node. If by doing so keeps the performance on the validation set at the same level or improves it, the removal is accepted otherwise it's not. The process is repeated for all the candidate nodes.

c) How does a random forest output a decision for classification and regression?
Ans: For classification, the mode of the predicted labels is considered. For regression, we may average the value over all the decision trees.